



## AQ11.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Let  $f(x, y) = x \log(y)$  and  $Hf(x, y)$  denote the Hessian matrix of  $f$  at  $(x, y)$ . Which of the following options are correct?

☐

Rank of  $Hf(0, 1)$  is 2.

☐

Rank of  $Hf(0, 1)$  is 1.

☐

$(0, 1)$  is a local minimum.

☐

$(0, 1)$  is a saddle point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Rank of  $Hf(0, 1)$  is 2.

$(0, 1)$  is a saddle point.

1 point

Choose the correct statements about  $f(x, y) = x^2 + axy + by^2$ .

☐

$(0, 0)$  is a saddle point of  $f$  iff  $4b - a^2 < 0$ .

☐

$(0, 0)$  is a saddle point of  $f$  iff  $4b - a^2 > 0$ .

☐

There are infinitely many possible pairs  $(a, b)$  for which  $f$  has local minimum at  $(0, 0)$ .

☐

If  $a = b = 1$  then  $f$  has a local maximum at  $(0, 0)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, 0)$  is a saddle point of  $f$  iff  $4b - a^2 < 0$ .

There are infinitely many possible pairs  $(a, b)$  for which  $f$  has local minimum at  $(0, 0)$ .

1 point

Consider the following function

$$f(x, y) = x^2 + xy + y^2 + 2x - 2y + 5$$

Which of the following option(s) is (are) true?

☐

$(-2, 2)$  is a critical point of  $f$ .

☐

$(2, -2)$  is a critical point of  $f$ .

☐


Determinant of the Hessian matrix at point  $(x, y)$  is 33.

☐

$f$  has a local minimum at point  $(-2, 2)$ .

☐

$(2, -2)$  is a saddle point of  $f$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(-2, 2)$  is a critical point of  $f$ .

Determinant of the Hessian matrix at point  $(x, y)$  is 33.

$f$  has a local minimum at point  $(-2, 2)$ .

For what value of  $k$  is the Hessian test inconclusive for the function  $f(x, y) = x^3 + kxy + y^3$  at  $(0, 0)$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

## Level 2

1 point

Choose the correct statements about  $f(x, y) = 2x^3 - 3x^2y - 12x^2 - 3y^2$

☐

$f$  has 33 critical points.

☐

$(0, 0)$  is a local maximum.

☐

$(0, 0)$  is a saddle point.

☐

$(-4, -8)$  is a saddle point.

☐

$(-4, -8)$  is a local minimum.

☐

$f$  has a global minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f$  has 33 critical points.

$(0, 0)$  is a local maximum.

$(-4, -8)$  is a saddle point.

1 point

Choose the correct statements about  $f(x, y) = x^8 - 2x^4y + y^3 - y$

☐

The Hessian matrix of  $f$  at point  $(x, y)$  is  $\begin{bmatrix} 56x^6 - 24x^2 & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$

☐

The Hessian matrix of  $f$  at point  $(x, y)$  is  $\begin{bmatrix} 56x^6 - 24x^2y^8x^3 & 6y \\ 8x^3 & 6y \end{bmatrix}$ .

☐

$f$  has four critical points.

☐

$(1, 1)$  is a point of local minimum of  $f$ .

☐

$(1, 1)$  is a point of local maximum of  $f$ .

☐

Second derivative test is inconclusive for the point  $(0, \frac{1}{\sqrt{3}})$ .

☐

1).

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The Hessian matrix of  $f$  at point  $(x, y)$  is  $\begin{bmatrix} 56x^6 - 24x^2y^8x^3 & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$ .

$f$  has four critical points.

$(1, 1)$  is a point of local minimum of  $f$ .

Second derivative test is inconclusive for the point  $(0, \frac{1}{\sqrt{3}})$ .

☐

1).

If  $M$  denotes the local maximum value of the function  $F(a, b) = \int_a^b (3 - 2x - x^2) dx$  over all possible values of  $a$  and  $b$  then find the value of  $3M$ . Note that the function has only one local maximum and you have to find that.

3 - 2x - x<sup>2</sup> dx over all possible values of  $a$  and  $b$  then find the value of  $3M$ . Note that the function has only one local maximum and you have to find that.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 32

1 point

1 point

Consider a function  $f(x, y)$ . Let  $v_1, v_2, v_3$  be the critical points of the function  $f$  and  $H_k$  be the Hessian matrix of the function at the point  $v_k$ . Then which of the following options is(are) true?

☐

If  $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$ , then  $v_1$  is a saddle point.

☐

If  $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$ , then  $v_2$  is a saddle point.

☐

If  $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$ , then  $v_3$  is a point of local extremum.

☐

If  $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ , then  $v_1$  is a point of local minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$ , then  $v_2$  is a saddle point.

If  $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$   $H_3 = [3-311]$ , then  $v_3$   $v_3$  is a point of local extremum.

If  $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$   $H_1 = [2002]$ , then  $v_1$   $v_1$  is a point of local minimum.

Check Answers

Your score is: 0/8